

Clinical Anchored: Strategy as a Game

Last updated: 2026-09-23. Working model, round 3: updated with Sarah's email of Sept 22, plus a mathematical appendix. Payoffs are ordinal and illustrative; trust the conclusions that do not depend on the numbers. Not legal advice. The PDF for Sarah ("Patient Messaging Tool — Strategy as a Game") matches this doc, minus the Superseded note, plus a sources list.

What Sarah's Sept 22 email settled

- No information from the hospital chart or ConnectingOntario goes into the app. ConnectingOntario is for reading and gathering only; lifting information from it is not allowed. Including hospital information at this stage would be a major privacy breach.
- Patient-reported answers are in: patients answer questions themselves, which avoids the chart-privacy issue.
- v1 focus: post-op patients. A dropdown of the procedures she performs, and checkboxes rather than free-form answers, because many patients struggle to describe symptoms. Wound photo sharing would be valuable later.
- Her own clinic records (Accuro) hold the most important information; her clinic notes matter most for dealing with patients.
- The pain: post-op phone calls despite detailed written instructions. Goals: fewer calls to the office, fewer callbacks she has to make, fewer follow-up clinic visits, so she can focus on new consults and sicker patients.
- Path: a tool for her first, then her colleagues, mainly as an outpatient tool for clinic patients. If it ever goes to the hospital, the angle is preventing emergency department visits.
- She is not involving Mackenzie Health admin.

Objective and constraint

max over strategy σ : $E[w_1 \cdot \text{paying partners}(\sigma) + w_2 \cdot \text{capital raised}(\sigma)] - \text{Drag}(\sigma)$
subject to: $\text{Legal}(\sigma) \geq \text{Floor}(\text{scale reached})$

- **Drag decomposes by severity:** veto exposure (can end the game) $\gg$ control rights ceded (IP, data, publication, timeline) $>$ delay. Minimize in that order.
- **The legal floor is set by the success state, not today's state.** Detection probability rises with scale (patients, physicians, investor diligence), so comply to the scaled standard from day one. Compliance debt is priced by investors at diligence.
- **Product scope moves the floor.** A platform serving two or more custodians primarily to let them disclose PHI to one another is a health information network provider (HINP) under O. Reg. 329/04 s. 6, and must do and hand over a written threat/risk and privacy assessment. PIAs are otherwise not formally required under PHIPA. Keep practices as separate tenants with no custodian-to-custodian sharing to stay out of HINP; still do a light PIA as a signal.
- **Penalties are built so that shortcuts never pay.** IPC administrative monetary penalties in force since Jan 1, 2024: up to \$50K for individuals, \$500K for organizations, and can exceed caps to strip economic benefit. The criteria include preventative steps not taken and economic benefit gained.

Solution concepts

Sequential moves with private information, so plain Nash admits non-credible threats. Use subgame perfection (backward induction) for sequencing, perfect Bayesian equilibrium for signaling to investors and physicians, and veto-player logic for drag.

Players

Player	Moves	Veto?	Wants
Ian + Sarah (founders)	Scope, disclosure, sequencing, signals	n/a	Partners and capital, low drag
Sarah (separately)	Participate / exit	Over everything (participation constraint)	Time back, fewer calls and follow-ups, care quality, no risk to license, privileges or board standing
Mackenzie admin	Acknowledge / assert jurisdiction	Only inside jurisdiction: hospital systems, data, resources, branding; her conflict if she is a director	Low effort, no risk ownership, no embarrassment, good relations with medical staff leadership
Ontario Health (ConnectingOntario)	Audit, suspend or revoke access	Out by design: no ConnectingOntario data in the app	Use only as authorized
IPC, CPSO, Health Canada	React to complaints or triggers	Legal, when triggered	Compliance
Patients	Consent, answer check-ins, message	Over their own participation	Reassurance, fewer unnecessary visits
Physicians	Adopt and pay / not	Over revenue	Time saved at acceptable risk and price
Investors (incl. IRAP)	Fund / pass	Over capital	Signals they can verify
REB, evaluator, institutions	Only if invited	Opt-in	Their own process
Competitors (Epic, AI scribe vendors, SeamlessMD)	Enter / ignore	No	Protect or extend their footprint

Veto set after her email: {Sarah, the law}. Mackenzie admin leaves the veto set through scope design; Ontario Health leaves because the app never touches its data. Opt-in players stay out until their signal value exceeds their drag.

Subgame 1: Mackenzie admin (disclosure)

Payoffs (founders, admin); p = probability admin learns of it other than from her.

Founders' move	Admin acknowledges	Admin asserts jurisdiction
Ask permission	(2, 1): approval makes admin co-own the risk	(1, 2) admin best response

Founders' move	Admin acknowledges	Admin asserts jurisdiction
Inform, with scope boundaries	(3, 3) admin best response	(2, 1)
Stay quiet, discovered	(4, 0)	(0, 2) admin best response

Undiscovered quiet pays (4, n/a) with probability $1 - p$.

- Asking concedes jurisdiction, and approval transfers risk to admin, so their best response is a full review. Asking is dominated whenever admin lacks jurisdiction. This supports her decision not to involve admin.
- Asserting against a disclosed, out-of-jurisdiction private-practice tool costs admin more than acknowledging. The threat is not credible.
- **Staying quiet is a Nash equilibrium, not a subgame-perfect one.** (Quiet; admin asserts everywhere) is Nash for $p \leq 1/2$, but rests on the off-path threat to assert after being informed. Under subgame perfection, inform (3) beats quiet ($4(1 - p)$) once $p > 1/4$.
- p is endogenous: success drives it toward 1 (colleagues, patients, investor diligence about her hospital role). Quiet is optimal only in the world where the project stays small.
- Quiet converts a non-credible threat into a credible one. If she is a director, an undisclosed family interest hands admin a governance hook regardless of the project's scope.
- The concern about admin is correct for anything inside their jurisdiction, which is why scope design comes first. Her email already draws that line (no hospital chart or ConnectingOntario data).

Equilibrium: keep the hospital out of the approval loop; if she is a director, a short written disclosure of the family interest (information, not a request); admin acknowledges.

Subgame 2: Ontario Health (resolved by exclusion)

- Inspection-game logic: the gain from shortcuts is small and the penalty (loss of ConnectingOntario access, IPC penalties, reputation) is catastrophic, so full compliance is dominant.
- Resolved Sept 22: Sarah confirms lifting information from ConnectingOntario is not allowed. The Ontario Health Acceptable Use Policy (v1.1, June 2020) likewise limits use "solely for the purposes authorized in the applicable agreement" and allows suspension or revocation.
- The chart-import stage via ConnectingOntario is dropped. Ontario Health leaves the game. The chart-summary value moves to her own Accuro records in a later stage.

Rule: **ask only players who already hold a veto; inform those who don't; design out the rest; invite opt-in players only when their signal is worth more than their drag.**

Subgame 3: Investors (signaling, least-cost separating)

Founders know product quality; investors don't. A signal separates only if a weak product cannot cheaply produce it. In the least-cost separating (Riley) outcome the strong type sends the cheapest signal that clears the current investor's bar; anything more is waste, paid here in drag.

Signal	Can a weak product fake it?	Drag	Verdict
Sarah uses it	Yes	None	Necessary, weak signal

Signal	Can a weak product fake it?	Drag	Verdict
Her partners use it	Mostly	None	Weak
Outside physicians pay	No	None	Strong; best ratio
Retention over 6-12 months	No	None	Strong
Fewer post-op calls, callbacks and follow-up visits vs pre-registered baseline	Hard, if pre-registered and tamper-evident	None	Strong; these are the outcomes she named
IRAP award	Hard (ITA screens)	Low	Moderate
Endorsements (Mi2, OntarioMD)	Yes; endorser commits nothing	High	Cheap talk; skip
Independent evaluation (academic + REB)	No	Medium-high	Strongest; priced for Series A
Institutional buyer validation (EAHN, CAN Health)	No	High	Strong; hospital path only (ED-visit angle)

Mechanism-design move: before launch, hash and anchor the metric definitions (calls to the office, callbacks, follow-up visits; later ED visits), the analysis plan, a baseline summary and an all-users reporting commitment. The hash-chained log plus periodic anchored roots makes post-hoc metric selection and post-hoc data alteration detectable. This raises the separating power of free signals and lets the paid independent evaluator wait until the Series A bar requires it. Only hashes of plans and aggregates go on-chain.

The same log covers what PHIPA s. 10.1 would require of electronic audit logs (type of information, time, who accessed, whose record). Enacted 2020; pending regulations per the sources found; check current status.

Subgame 4: Physicians (coordination)

- Within a practice, the secretary relay disappears only if the whole practice switches: a stag hunt. All-adopt is payoff-dominant; none-adopt is risk-dominant when the benefit is less than twice the cost (appendix A.5). Select the good equilibrium by selling to the practice as a unit (one decision, one PHIPA agreement) with a champion. Her plan to disseminate to colleagues once it works fits this.
- Outside physicians do not adopt a tool that lacks signed agent agreements, patient consent and CMPA-consistent use. Legal compliance is a precondition of the adoption equilibrium, not just a constraint.
- A returning messaging billing code would shift physician payoffs sharply. Watch it.

Sarah's participation constraint

Her utility: time saved (fewer calls, callbacks, follow-ups) + upside (if equity) + care quality – professional risk – governance risk – time. Any strategy with a real chance of costing her privileges, license or board standing violates the constraint and ends the game. Her email satisfies it by construction: nothing from the hospital, patient-reported data only. Equity strengthens her commitment and the investor read (committed clinical cofounder) but raises conflict-of-interest weight. She frames it as "a tool meant for me at the moment", so the venture objective and her role need explicit alignment before investor steps.

Competitors

AI scribe vendors have the lowest cost of entering this niche; Epic's model and buyer are elsewhere; SeamlessMD overlaps with structured post-op check-ins but sells to hospitals, not private practices. Best response: speed, switching costs (practice-level workflow, procedure and protocol library, attested history, Accuro integration), and keep partnership or exit optionality.

Equilibrium path (backward induction)

1. Choose the game: separate entity; private-practice scope; no hospital systems, data, resources or branding; no hospital chart or ConnectingOntario data; separate tenants (not a HINP); red-flag rules support rather than replace her judgement.
2. If she is a director: short written disclosure of the family interest to Mackenzie (information, not a request). Before any real patient: patient consent, a written privacy agreement between her clinic and the company, Canadian hosting.
3. IRAP and SR&ED. Measure the baseline (post-op calls, callbacks, follow-up visits); pre-register and anchor the metrics.
4. v1 with her post-op patients: procedure dropdown, checkbox check-ins, messaging.
5. Her colleagues at the practice, as a unit.
6. Outside practices, paying. Later: wound photos; Accuro notes.
7. Raise on least-cost separating signals.
8. Choose the buyer path from the data. Hospital path uses the ED-visit angle; independent evaluator + REB for the next bar; path-specific stakeholders on their triggers.

Deviation check: admin acknowledging beats asserting; Ontario Health is out of the game; other monitors have nothing to act on; physicians adopt once compliant with proof and ROI; investors fund on separating signals; Sarah stays because the path is safe for her. Founders would not deviate to quiet (dominated once $p > p^*$), to asking (concedes jurisdiction), or to early stakeholders (overpays for signals, adds veto and control exposure).

Parameters that can flip the equilibrium

- Her board status: if a current director, disclosure is a formal conflict obligation, not just strategy.
- Her equity and the venture objective: she frames the tool as hers for now; align on intent and role.
- Free text vs structured: she prefers patients avoid free-form answers; the current decision keeps free text alongside dropdowns. Structured-first also limits what sensitive detail enters the system.
- Investor bar: if Canadian health investors require institutional validation earlier, hospital-path stakeholders move up.
- Billing code return: shifts physician payoffs; get Ontario Health verification.
- AI scribe vendor entry: raises the speed premium; consider partnership.

Appendix: mathematical details

Each result is stated in general form first and with the illustrative payoffs second. Expected-value comparisons treat the illustrative payoffs as cardinal; the best-response results need only their ordering.

A.1 Objective, drag and the legal floor

$$\max \text{ over } \sigma \in \Sigma: U_F(\sigma) = E[w_1 \cdot N(\sigma) + w_2 \cdot K(\sigma)] - D(\sigma), \text{ subject to } \Lambda(\sigma) \geq \bar{\Lambda}(n(\sigma))$$

N = paying partners, K = capital raised, D = drag, Λ = legal compliance, $\bar{\Lambda}$ = the legal floor, non-decreasing in scale n .

$$D(\sigma) = \sum_j [v_j(\sigma) \cdot V + c_j(\sigma) + \delta_j(\sigma)]$$

v_j = probability that player j uses a veto; V = value of the whole project; c_j = control rights ceded (IP, data, publication, timeline); δ_j = cost of delay. Because V is everything at stake, $v_j \cdot V$ dominates for any non-negligible v_j : veto exposure first, then control, then delay.

A compliance gap that saves g and costs P if detected has expected value $g - \pi(n) \cdot P$, where the detection probability $\pi(n)$ rises with scale and approaches 1 at investor diligence. The binding floor is the one at the target scale.

A.2 Subgame 1: disclosure to hospital admin

Moves: founders F choose A (ask), I (inform, with scope boundaries) or Q (quiet). After A or I , admin H chooses k (acknowledge) or a (assert). After Q , nature reveals the project to H with probability p ; if revealed, H chooses k or a ; if not, the game ends.

Founders' payoffs: $q_{hi} = 4$ (quiet, never revealed), $q_{lo} = 0$ (quiet, revealed, admin asserts), $i = 3$ (inform, acknowledged), $i_a = 2$ (inform, asserted), $\alpha = 1$ (ask, asserted).

Backward induction: $BR_H(A) = a$ ($2 > 1$), $BR_H(I) = k$ ($3 > 1$), $BR_H(Q, \text{revealed}) = a$ ($2 > 0$). Continuation values: $V(A) = \alpha = 1$, $V(I) = i = 3$, $V(Q) = (1 - p) \cdot q_{hi} + p \cdot q_{lo} = 4(1 - p)$.

- **Proposition 1 (SPE).** Inform iff $p \geq p^* = (q_{hi} - i)/(q_{hi} - q_{lo}) = (4 - 3)/(4 - 0) = 1/4$. Asking is never optimal, since $\alpha < i$. The numerator $q_{hi} - i$ is the cost of disclosing; cheap disclosure means a small p^* .
- **Proposition 2 (non-credible threat).** (Quiet; assert at every information set) is a Nash equilibrium iff $p \leq (q_{hi} - i_a)/(q_{hi} - q_{lo}) = (4 - 2)/4 = 1/2$: given admin asserts everywhere, $V(I) = 2$, and $V(Q) = 4(1 - p) \geq 2$ exactly when $p \leq 1/2$; admin's off-path choices cost it nothing. Not subgame perfect: after I , asserting gives admin 1 rather than 3. For $p \in (1/4, 1/2]$, Nash supports "stay quiet because they would block us" while SPE prescribes informing.
- **Proposition 3 (scale).** $p = p(n)$ increasing with $p(n) \rightarrow 1$ at the fundraise, so $V(Q) \rightarrow q_{lo} < i$. Quiet is optimal only if the project stays small.
- **Robustness, obstructive admin.** With probability θ admin is obstructive and asserts even after I : $V(I) = (1 - \theta) \cdot 3 + \theta \cdot 2 = 3 - \theta$. Inform $\geq$ quiet iff $p \geq (1 + \theta)/4$; $V(I) \geq 2 > \alpha = V(A)$ for all $\theta \in [0, 1]$. Even with $\theta = 1$, inform is optimal once $p \geq 1/2$, and asking stays dominated.
- **Robustness, director.** An undisclosed family interest adds governance cost $\gamma > 0$ to the revealed branch: $q_{lo} = -\gamma$, so $p^* = (q_{hi} - i)/(q_{hi} + \gamma) = 1/(4 + \gamma) < 1/4$.

A.3 Subgame 2: inspection games (Ontario Health and the regulators)

Founders choose comply (C) or shortcut (S). A shortcut saves $g > 0$ and costs P if detected. Detection comes from inspection (probability r of inspecting, d of finding) and independently from complaints and diligence, $\pi_0(n)$:

$$\pi(r, n) = 1 - (1 - \pi_0(n)) \cdot (1 - r \cdot d) \geq \pi_0(n)$$

Founders comply whatever the monitor does iff $g \leq \pi_0(n) \cdot P$. In the classic inspection game the monitor randomises because inspecting is costly and the inspectee is left indifferent. Here the condition holds even with no inspection once scale makes complaints and diligence likely, so compliance is strictly dominant and the equilibrium is pure: (comply, no inspection).

Magnitudes: g is small (weeks of setup). P includes loss of her ConnectingOntario access, IPC penalties up to \$50K per individual and \$500K per organization (more to remove any benefit), professional discipline, and investors walking away at diligence, where $\pi_0 \approx 1$. Excluding ConnectingOntario data removes the shortcut action for that monitor entirely.

A.4 Subgame 3: signalling to investors

Types $\theta \in \{H, L\}$ (strong or weak product), prior $\mu_0 = \Pr(H)$. Evidence $s \geq 0$ costs $c_\theta(s)$, with single crossing: $c_L(s) - c_H(s)$ increasing in s . An investor in round r funds iff the posterior $\mu(s) \geq \bar{\mu}_r$, with the bar rising by round. Funding is worth B .

Separating equilibrium (weak sends nothing, strong sends s): $B - c_L(s) \leq 0$ (weak won't mimic) and $B - c_H(s) \geq 0$ (strong is willing).

Least-cost separating outcome (selected by the intuitive criterion): $s^* = \min\{s : c_L(s) \geq B\}$. Evidence beyond s^* is waste, paid here in drag.

Several kinds of evidence j : separating power $\Delta_j = c_L^j - c_H^j$, cost to the strong type $c_H^j + d_j$ (d_j = drag). The efficient bundle reaches $c_L \geq B$ at minimum $\Sigma_j(c_H^j + d_j)$, so favour a high $\Delta_j/(c_H^j + d_j)$:

- Outside physicians paying, retention and pre-registered savings: large Δ_j , $d_j \approx 0$.
- Endorsements: $\Delta_j \approx 0$ (the endorser commits nothing; cheap talk).
- Independent evaluation: large Δ_j and large d_j ; efficient only when $\bar{\mu}_r$ rises (around Series A).

Pre-registration as a commitment device: without commitment a weak type can select or alter metrics after the fact at cost m , so $c_L(\text{metric}) \approx m$. Anchoring a hash of the metric definitions and analysis plan before launch, plus the hash-chained log, makes both detectable, pushing the effective m toward infinity with c_H unchanged. Δ_{metric} rises at almost no cost. The all-users reporting commitment covers selection of which practices to report.

A.5 Subgame 4: coordination within a practice

Two partners (extends to more) choose adopt (A) or not (N); the relay only disappears if both adopt. Benefit b , cost c (including expected legal risk):

	Partner 2 adopts	Partner 2 does not
Partner 1 adopts	$(b - c, b - c)$	$(-c, 0)$
Partner 1 does not	$(0, -c)$	$(0, 0)$

If $b > c$, both (A, A) and (N, N) are Nash equilibria; (A, A) is payoff-dominant, and risk-dominant iff $b - c > c$, i.e. $b > 2c$. A single practice-level decision (one contract, one agreement) turns two choices into one and selects (A, A). With $c = c_0 + \rho \cdot L_{\text{phys}}$ (ρ = probability of a legal or professional problem, L_{phys} = the physician's loss), missing agreements and consent make $c > b$, and (N, N) is the only equilibrium.

A.6 Sarah's participation constraint

$$u_S(\sigma) = T(\sigma) + e \cdot W(\sigma) + Q(\sigma) - \rho_{\text{prof}}(\sigma) \cdot L_{\text{prof}} - \rho_{\text{gov}}(\sigma) \cdot L_{\text{gov}} - \tau(\sigma) \geq \bar{u}_S$$

T = time saved, e = equity share, W = venture value, Q = care quality, ρ = probabilities of professional and governance harm with large losses L , τ = her time, $\bar{u}_S$ = status quo. Large L means the constraint effectively requires $\rho_{\text{prof}} \approx \rho_{\text{gov}} \approx 0$: a feasibility restriction on plans rather than a trade-off. Her email's rules keep ρ_{prof} near zero for data handling. Equity raises $e \cdot W$ (commitment, investor read) and the conflict-of-interest weight.

A.7 Veto players and when to invite a stakeholder

$\Pr(\text{survive}) = \Pi$ over $j \in \mathcal{U}$ of $(1 - v_j)$, which falls strictly with every veto player added with $v_j > 0$. Scope design sets $v_H = v_O = 0$, leaving $\mathcal{U} = \{\text{Sarah, the law}\}$. Invite opt-in stakeholder j iff $\Delta B_j \geq v_j \cdot V + c_j + \delta_j$. Before usage, $\Delta B_j \approx 0$ for most j ; the rule admits the independent evaluator once the investor bar rises and results exist.

A.8 Sequencing: irreversibility and the value of waiting

Savings estimate: $\hat{\Delta} = \bar{Y}_{\text{post}} - \bar{Y}_{\text{pre}}$, with $\bar{Y}$ the average post-op calls, callbacks and follow-up visits per patient. With a comparator surgeon C , difference-in-differences: $\hat{\Delta}_{\text{DiD}} = (\bar{Y}_{\text{post}}^T - \bar{Y}_{\text{pre}}^T) - (\bar{Y}_{\text{post}}^C - \bar{Y}_{\text{pre}}^C)$. $\bar{Y}_{\text{pre}}$ is only observable before launch, so the baseline is the one move that cannot be postponed.

Buyer path: deferring the choice between practices (V_A) and institutions (V_B) until data arrive is worth $E[\max(V_A, V_B)] - \max(E[V_A], E[V_B]) \geq 0$ (convexity of max, Jensen's inequality).

A.9 What is robust and what is not

- Robust to the numbers: admin's best responses and hence inform over ask; compliance dominance at scale; least-cost signalling and the value of pre-registration; survival falling with each veto player; the value of waiting on the buyer path.
- Parameter-dependent: p^* , the obstructive share θ , whether $b > 2c$ within a practice, and the investor bars $\bar{\mu}_r$.

Superseded

Round 2 modeled "no patient data at all", from a paraphrase of her email. The actual constraint is narrower (no hospital chart or ConnectingOntario data; patient-reported data is in), which preserves the strong usage signals.